

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 20

Code of Conduct

I **MANU NEETHI S (192521063)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **MANU NEETHI S**

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

QUESTION 1

1. Problem Analysis

A university operating campuses in multiple countries needs its students, faculty, and administrative staff to access shared academic resources — Learning Management Systems (LMS), research computing clusters, digital libraries, and student information systems — regardless of which campus or country they are physically located in. Building one massive centralized cloud is inefficient because of data-residency laws (such as GDPR in Europe), high cross-border latency, and the fact that each campus may already have invested in a private or regional public cloud. Cloud federation solves this by interconnecting these independent cloud environments into a single, cooperatively managed system while letting each campus retain administrative and regulatory control over its own infrastructure.

2. Design Objectives

- Seamless resource sharing: a student in one country can use compute/storage hosted at another campus without manual provisioning.
- Data sovereignty compliance: sensitive student and research data stays within the legal jurisdiction it was created in.
- High availability and load balancing across campuses, especially during peak periods (e.g., exam result publishing, semester registration).
- Unified identity so one login works everywhere (single university-wide credential).
- Cost optimization by bursting workloads to partner-campus or public-cloud capacity only when local resources are exhausted.

3. Proposed Federated Cloud Architecture

The recommended design is a hybrid, broker-mediated cloud federation, structured in four layers:

Layer 1 – Campus Cloud Nodes: Each campus (e.g., India, UK, Malaysia, UAE) operates its own private/community cloud built on a virtualization layer (VMware vSphere / OpenStack) hosting local LMS instances, research VMs, and student databases. This node handles day-to-day local workloads with minimum latency.

Layer 2 – Federation Gateway / Cloud Broker: A dedicated broker service (e.g., built using open standards such as OCCI or a custom API gateway) sits at each campus boundary. It authenticates federation requests, translates resource requests between heterogeneous cloud platforms, enforces Service Level Agreements (SLAs) between campuses, and performs intelligent workload routing (send a request to the nearest available node with capacity).

Layer 3 – Federated Identity and Access Management (IAM): A central Identity Federation service, using SAML 2.0 or OAuth 2.0/OIDC with a protocol such as Shibboleth/eduGAIN (already widely used by universities), issues a single set of credentials trusted by every campus. This gives Single Sign-On (SSO) across all federated nodes without duplicating user directories.

Layer 4 – Shared Resource Pool and Data Fabric: A distributed, replicated storage/content-delivery layer (e.g., an object store replicated with policy-based geo-fencing) holds shared academic content — MOOC videos, research datasets, digital library assets — cached at the nearest campus edge node, while campus-specific student records remain in local, non-replicated storage to satisfy data-residency rules.

4. Workflow Example

When a student enrolled at the India campus travels to the UK campus for a semester exchange, she logs in using her university-wide SSO credential. The UK campus's federation gateway recognizes the federated identity token issued by the India campus's IAM, grants her access to local compute labs and the shared LMS, and her academic records remain securely stored at the India campus, only queried on demand through the federation gateway rather than being copied wholesale — thereby balancing convenience with data-governance requirements.

5. Key Design Justifications

- Broker-based federation (rather than a single global cloud) keeps each campus's existing infrastructure investment usable and avoids a single point of failure.
- Federated IAM eliminates duplicate account management overhead across countries and improves security auditability.
- Policy-based data placement (keep-local vs. replicate-globally) ensures compliance with region-specific data protection laws.
- SLA-driven resource bursting allows a campus experiencing a load spike (e.g., during result declaration) to borrow idle compute capacity from a partner campus in a different time zone.

6. Expected Benefits

- Improved resource utilization — idle capacity at one campus serves demand spikes at another.
- Lower latency for local users since most requests are served from the nearest campus node.
- Regulatory compliance through data localization policies enforced at the federation-gateway layer.
- Business continuity — if one campus's cloud suffers an outage, workloads can be redirected to a federated partner.

Question 2

1. Conceptual Distinction

Although both cloud federation and hybrid cloud combine more than one infrastructure environment, they differ fundamentally in ownership, control, and purpose. Cloud federation is a horizontal partnership between two or more independent, typically peer, cloud providers or organizations that agree to interconnect and share resources under mutually negotiated policies — as in the university example above, where campuses in different countries, each owning its own cloud, cooperate as equals. Hybrid cloud, by contrast, is a vertical integration within a single organization, combining its own private cloud (or on-premises data center) with one or more public cloud providers, all under one unified administrative control.

2. Comparative Analysis

Ownership	Multiple independent organizations/providers, each retaining control of its own cloud	Single organization owning/controlling both private and public segments
Relationship type	Peer-to-peer partnership governed by federation agreements/SLAs	Internal integration governed by one organization's own IT policy
Primary goal	Resource sharing, interoperability, and mutual capacity bursting between separate entities	Flexibility to run sensitive workloads on-premises while scaling elastic workloads to public cloud
Identity management	Federated IAM (SSO trust established across organizations, e.g., SAML/OIDC federation)	Centralized IAM within the same organization, extended to the public cloud via connectors
Data governance	Complex — must reconcile differing policies of multiple legal entities/jurisdictions	Simpler — one governance policy applied consistently across both environments
Typical use case	University campuses, research consortia, multi-provider disaster recovery pacts	An enterprise keeping customer databases on-premises while running web front-ends on AWS/Azure
Trust model	Explicit trust/federation agreements between separate administrative domains	Implicit — both sides belong to the same administrative domain

Interoperability challenge	High — providers may use different platforms, APIs, and standards	Lower — usually managed via a single vendor's hybrid tooling (e.g., Azure Arc, AWS Outposts)
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3. Illustrative Example

In the university scenario, if the India campus's private OpenStack cloud federates with the UK campus's VMware-based private cloud under a mutual resource-sharing agreement, that is cloud federation. If, instead, the India campus alone decided to keep its student database on its own private servers but moved its video-lecture streaming workload to a public cloud such as AWS for elastic scaling during peak viewing hours, that would be a hybrid cloud deployment — a single organization spanning private and public infrastructure.

4. Conclusion

Cloud federation is best understood as "cloud-to-cloud cooperation across organizational boundaries," prioritizing interoperability and shared governance among peers, whereas hybrid cloud is "infrastructure diversification within one organization," prioritizing flexibility and cost-efficiency under unified control. A large multinational university system may, in fact, use both simultaneously: each campus runs a hybrid cloud internally (private + public), and the campuses then federate with one another to form the larger, multi-country academic cloud described in Question